

Free Will as Structural Selection: A TNA-Based Resolution Beyond Determinism and Randomness

Abstract

The classical problem of free will is traditionally framed as a dichotomy between determinism and indeterminism. In deterministic systems, all future states are uniquely determined by initial conditions, leaving no room for genuine agency. In indeterministic frameworks, outcomes are fundamentally random, undermining the notion of controlled choice. Both perspectives fail to provide a satisfactory account of free will.

In this work, we propose a structural reformulation of the problem using the TNA (Theory of Necessary Axioms) framework. We argue that free will emerges in dynamical regimes characterized by non-uniqueness of solutions, where identical initial conditions admit multiple admissible trajectories ($\Delta > 1$). In such regimes, the system does not violate physical laws but operates within regions where those laws fail to enforce a unique continuation.

We introduce the concept of free will as a selection process over structurally admissible trajectories and distinguish this from both deterministic evolution and ontological randomness. We further argue that non-computability plays a critical role in preventing the reduction of such selection processes to algorithmic procedures. This framework provides a unified interpretation of agency compatible with physical law, while avoiding both strict determinism and pure randomness.

1. Introduction

The question of free will has long been constrained by a conceptual dichotomy: either the universe is deterministic, in which case all actions are predetermined, or it is indeterministic, in which case actions are random. In both cases, genuine agency appears impossible.

This paper challenges that dichotomy. We propose that the problem arises from an incomplete structural characterization of dynamical systems. Specifically, the classical debate ignores regimes in which the evolution of a system is not uniquely defined by its initial conditions.

Using the TNA framework, we reinterpret free will as a phenomenon emerging from structural non-uniqueness and constrained selection.

2. The Classical Deadlock

2.1 Determinism

In classical mechanics, the state of a system evolves according to differential equations that satisfy Lipschitz continuity. Under these conditions, the Picard–Lindelöf theorem guarantees a unique solution for each initial condition.

Thus, for any initial state x_0 , there exists a single trajectory:

$$|\Delta(x_0)| = 1$$

This implies that all future states are fixed, eliminating the possibility of alternative outcomes.

2.2 Indeterminism

In contrast, certain interpretations of quantum mechanics introduce fundamental randomness. Outcomes are not determined but sampled from probability distributions.

However, randomness does not restore agency. A random outcome is not a chosen outcome.

2.3 The Failure of the Dichotomy

Both determinism and indeterminism eliminate meaningful agency:

- Determinism removes alternatives
- Randomness removes control

A new framework is required.

3. Structural Reframing via TNA

We consider a dynamical system:

$$[\dot{x} = f(x), \quad x \in \Omega]$$

Define:

- $\Delta(x_0)$: set of admissible trajectories
- $F_{\min}$: minimal structural constraint ensuring uniqueness

3.1 Structural Closure

If $F_{\min} < \infty$ (Lipschitz domain):

$$|\Delta(x_0)| = 1$$

The system is structurally closed.

3.2 Structural Breakdown

If $F_{\min} \rightarrow \infty$ (non-Lipschitz domain):

$$|\Delta(x_0)| > 1$$

Multiple trajectories are admissible.

This regime introduces **ontological multiplicity**, not randomness.

4. Non-Uniqueness as Ontological Freedom

In non-Lipschitz regimes, the system admits multiple valid continuations from the same initial condition.

This does not violate physical law. Rather, it reflects a limitation of structural constraints.

▮ *The system is not free from laws; it is free within them.*

This provides a necessary condition for free will:

$$|\Delta(x_0)| > 1$$

5. The Selection Problem

Multiplicity alone is insufficient for agency. A mechanism is required to select among admissible trajectories.

We define:

Free will as a selection process over $\Delta(x_0)$

Examples of selection mechanisms include:

- Filippov solutions in discontinuous systems
- Regularization in physical models
- Hypothetical quantum-gravitational collapse (Penrose)

The selection is constrained but not predetermined.

5A. Selection as Structural Constraint (N_1)

5A.1 From Mechanism to Constraint

The notion of “selection” introduced in Section 5 should not be interpreted as an additional dynamical mechanism acting within the same level as the system’s equations. Such an interpretation collapses agency into either deterministic resolution (e.g., Filippov rules) or stochastic choice (e.g., probabilistic collapse).

We instead reinterpret selection as a **higher-order structural constraint**, denoted N_1 , acting over the admissible set of trajectories $\Delta(x_0)$.

Formally:

- N_0 : dynamical level generating $\Delta(x_0)$
- N_1 : constraint level selecting within $\Delta(x_0)$

Thus, selection is not a trajectory, nor a rule applied post hoc, but a restriction intrinsic to the system’s structure.

5A.2 Agent as Constraint

Within this framework, the agent is not an element of $\Delta(x_0)$, but the structure that constrains it.

Agency = structural constraint over admissible trajectories

Let S denote the selection induced by N_1 . Then:

$$S : \Delta(x_0) \rightarrow \{\gamma\}$$

where γ is the realized trajectory.

Crucially, S is not reducible to a function defined solely on N_0 . It depends on global structural properties of the system.

5A.3 Non-Reducibility and Non-Computability

Because N_1 operates as a constraint over a set of admissible trajectories rather than as a step-by-step evolution rule, it need not be computable.

This sharpens the claim of Section 6:

- Non-computability is not merely epistemic (lack of algorithm)
- It reflects that selection is not generated by any local rule over N_0

Thus, unpredictability arises from **structural irreducibility**, not randomness.

5A.4 Ownership and Internal Constraints

A trajectory γ is “owned” by the system if it is selected under constraints generated internally by N_1 .

These constraints may encode:

- memory
- value structures
- coherence conditions across time

Without committing to a specific physical implementation, we define ownership structurally:

■ *A trajectory is an action if it results from internally generated constraints acting on $\Delta(x_0)$.*

5A.5 Relation to the Principle of Sufficient Reason

The TNA framework avoids both extremes:

- Deterministic sufficiency: a unique cause fixes the outcome
- Arbitrary selection: no reason at all

Instead, selection arises from a **configuration of constraints** that does not collapse $\Delta(x_0)$ to a singleton a priori.

Thus:

Reasons do not determine a unique outcome; they structure the space of admissible outcomes.

5A.6 Reformulated Definition of Free Will

We refine Section 7:

A system exhibits free will if:

1. Structural multiplicity exists: $|\Delta(x_0)| > 1$
 2. A higher-order constraint N_1 selects a trajectory
 3. The constraint is internally generated and not reducible to N_0 dynamics
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5A.7 Conceptual Consequence

This resolves the selection problem:

- Selection is not an unexplained mechanism
- It is a structural operation at a higher level

Thus:

The agent is the constraint that makes a trajectory actual among many admissible ones.

6. Non-Computability and Irreducibility

Even when a unique trajectory is realized, the selection process may not be computable.

If the system involves non-computable dynamics, no algorithm can predict the outcome in advance.

This prevents reduction to deterministic computation while avoiding randomness.

The decision is not random, but not algorithmically reducible.

7. TNA Definition of Free Will

We define free will within the TNA framework as follows:

A system exhibits free will if:

1. Structural multiplicity exists:
 $|\Delta(x_0)| > 1$
2. A selection mechanism operates:
 $\Delta \rightarrow \{\text{single trajectory}\}$
3. The selection is not reducible to deterministic computation

The selection process should not be interpreted as an additional dynamical mechanism. Rather, it corresponds to a higher-order structural constraint imposed by the system on the admissible set $\Delta(x_0)$.

In this sense, the agent is not a trajectory within Δ , but the structure that constrains Δ .

Agency thus emerges not from the generation of possibilities, but from their structured restriction.

8. Regimes of Agency

We classify systems according to their structural properties:

REGIME	Δ	FREE WILL
Lipschitz	1	No

REGIME	Δ	FREE WILL
Chaos	1	No
Non-Lipschitz	>1	Potential
Selection	$>1 \rightarrow 1$	Yes
Non-computable	1 (inaccessible)	Strong

Free will emerges only in regimes where multiplicity and selection coexist.

9. Implications

Physics

Randomness may reflect unresolved structural multiplicity rather than fundamental indeterminism.

Neuroscience

Decision-making systems may operate near structural boundaries where non-uniqueness arises.

Philosophy

Free will is compatible with physical law when reinterpreted structurally.

9A. Relation to Existing Theories of Agency (Kane, Dennett, List)

The TNA framework can be positioned relative to existing philosophical accounts of free will, clarifying both its points of convergence and its departures.

9A.1 Kane: Indeterminism and Self-Forming Actions

Robert Kane’s libertarian model locates free will in indeterministic neural processes, particularly in “self-forming actions” where multiple outcomes are genuinely possible.

The TNA framework provides a structural grounding for this intuition:

- Kane's indeterministic branching corresponds to regimes where $|\Delta(x_0)| > 1$
- Self-forming actions occur near structural boundaries $\partial\Omega$

However, TNA departs from Kane by avoiding probabilistic indeterminism. The multiplicity arises from **non-uniqueness**, not randomness.

9A.2 Dennett: Compatibilism and Functional Agency

Daniel Dennett defends a compatibilist account in which free will is understood in terms of evolved capacities such as deliberation, prediction, and control.

From the TNA perspective:

- These capacities operate primarily within regions where $|\Delta(x_0)| = 1$ (Lipschitz regimes)
- They explain **functional control**, but not the existence of genuine alternatives

Thus, TNA agrees that such capacities are necessary for agency, but not sufficient for **free will in the strong sense** defined here.

9A.3 List: The Ability to Do Otherwise

Christian List argues that free will requires the genuine possibility of alternative actions, even within a physically determined framework.

TNA provides a formal realization of this requirement:

$$|\Delta(x_0)| > 1$$

This condition encodes the “ability to do otherwise” in structural terms, without appealing to modal metaphysics.

9A.4 Synthesis

The TNA framework can be summarized in relation to these approaches:

- Kane → identifies where multiplicity matters

- Dennett → explains functional control within trajectories
- List → articulates the necessity of alternatives

TNA integrates these insights by distinguishing three layers:

1. Structural multiplicity ($|\Delta| > 1$)
2. Functional capacities (deliberation, evaluation)
3. Structural selection (resolution of Δ)

Free will arises only when all three are present.

9A.5 Conceptual Position

The TNA account is best understood as a form of **structural incompatibilism**:

- It rejects strict determinism ($|\Delta| = 1$)
- It rejects randomness as a basis for agency
- It locates freedom in regimes of structured non-uniqueness

Thus:

Free will is neither purely physical nor purely metaphysical, but emerges at the boundary between dynamical law and structural constraint.

9B. Ontology of Structural Constraint (N_1)

The introduction of N_1 as a higher-order structural constraint raises an ontological question: what is the nature of N_1 ? Three interpretations are compatible with the present framework.

9B.1 Physical Realization

N_1 may correspond to physical structures, such as neural configurations, synaptic weights, or dynamical patterns in brain activity. In this view, constraints emerge from material organization and evolve over time through plasticity and interaction.

9B.2 Functional Organization

Alternatively, N_1 may be understood at the level of functional organization. Here, constraints correspond to cognitive processes such as evaluation, deliberation, and goal maintenance. These are not reducible to individual physical states, but to patterns of operation across them.

9B.3 Informational Structure

A third interpretation treats N_1 as an informational structure: a configuration of memory, values, and internal representations that constrain admissible trajectories. Under this view, N_1 is defined by its role in structuring possibilities rather than its physical substrate.

9B.4 Pluralistic Interpretation

The TNA framework remains deliberately neutral among these interpretations. N_1 can be realized physically, functionally, or informationally, provided it acts as a constraint over $\Delta(x_0)$.

This suggests a layered ontology in which:

- N_0 defines the space of possible trajectories
- N_1 defines constraints shaped by the system's history

9B.5 The Problem of Regress

A potential objection is that if N_1 selects over Δ , one may posit a higher-level N_2 selecting over N_1 , leading to infinite regress.

TNA avoids this by treating N_1 not as an object requiring further selection, but as a closure condition:

■ *N_1 is the minimal level at which internal constraints become self-referential.*

At this point, the system's structure constrains its own evolution without requiring an external selector.

9B.6 Causality and Teleology

N_1 does not introduce teleology in a classical sense. Instead, it encodes constraints that can be interpreted as goal-directed behavior at higher levels.

Thus:

- At N_0 : causality is efficient (dynamical laws)
- At N_1 : causality appears as constraint-based (goal-like but structurally grounded)

This allows the framework to incorporate purposive behavior without invoking external final causes.

9B.7 Summary

The ontology of N_1 remains open but constrained:

- It must act as a selector over Δ
- It must be internally generated
- It must not reduce to local N_0 dynamics

This flexibility allows TNA to interface with multiple domains, including neuroscience, cognitive science, and information theory.

10. Conclusion

The classical opposition between determinism and randomness fails to capture the conditions required for agency.

Free will emerges in systems where:

- multiple trajectories are structurally admissible
- a selection process resolves this multiplicity

Thus:

Free will is not the violation of physical law, but the operation of a system within regions where the law does not enforce a unique outcome.

Appendix A — A Minimal Life-as-Dynamical-System Model

A.1 Setup

We construct a minimal dynamical system that captures decision-making as structural non-uniqueness.

Let:

$$\dot{y} = \sqrt{|y|}$$

with initial condition:

$$y(0) = 0$$

This system is continuous but not Lipschitz at $y = 0$.

A.2 Non-Uniqueness

This system admits infinitely many solutions:

- The trivial solution:

$$y(t) = 0$$

- Delayed departure solutions:

$$y(t) = \begin{cases} 0 & t \leq T \\ \frac{1}{4}(t - T)^2 & t > T \end{cases}$$

for any $T \geq 0$.

Thus:

$$|\Delta(0)| = \infty$$

A.3 Interpretation as Life Trajectory

Interpret:

- $y(t)$: degree of commitment / activation of a life trajectory
- $y = 0$: indecision or latent state
- departure from 0: irreversible commitment

Then:

- Staying at $y = 0 \rightarrow$ no decision
 - Leaving at time $T \rightarrow$ decision occurs at T
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A.4 Structural Meaning

At the critical point $y = 0$:

- The system admits multiple admissible futures
- No law enforces when transition must occur

Thus, the system is in a **non-Lipschitz regime of structural freedom**.

A.5 Mapping to Free Will

We identify:

- Δ : set of possible decision times T
- Selection: realization of a specific T

Free will corresponds to:

$T \in \Delta$ selected but not predetermined

A.6 Extension: Competing Trajectories

A richer model introduces competition:

$$\dot{y}_1 = \sqrt{|y_1|}, \quad \dot{y}_2 = \sqrt{|y_2|}$$

with constraint:

$$y_1 \cdot y_2 = 0$$

Interpretation:

- y_1 : trajectory A (e.g., stay)
- y_2 : trajectory B (e.g., leave)

Only one can activate.

The system must select:

$$(y_1, y_2) \in \{(> 0, 0), (0, > 0)\}$$

A.7 Conceptual Result

This minimal model shows:

- Non-Lipschitz dynamics $\rightarrow$ multiplicity of futures
- Selection $\rightarrow$ realization of one trajectory

Thus:

A human decision can be modeled as the selection of a trajectory emerging from a structurally non-unique dynamical system.

Appendix B — Everyday Manifestations of Structural Non-Uniqueness

B.1 Motivation

A natural question arises: if free will requires structural non-uniqueness ($|\Delta| > 1$), when do real human systems operate in such regimes?

This appendix provides concrete examples translating non-Lipschitz dynamics into lived experience.

B.2 Threshold Decisions (Irreversible Life Bifurcations)

Certain decisions act as structural bifurcation points in a life trajectory:

- Having or not having a child
- Leaving a long-term relationship
- Emigrating vs remaining
- Changing career paths irreversibly

At these points, the system (agent + history + environment) is near a boundary $\partial\Omega$ where small internal variations produce radically different long-term trajectories.

Interpretation:

These are macroscopic analogues of non-Lipschitz points, where multiple futures are structurally admissible.

B.3 Moral Critical Points

Moments of high ethical weight exhibit similar structure:

- Confessing vs concealing a harmful action
- Acting violently vs refraining
- Intervening vs remaining passive in critical situations

These situations are characterized by:

- High sensitivity to internal state
- Irreversibility of outcome
- Absence of a uniquely enforced trajectory

Interpretation:

The system operates in a regime where structure permits multiple coherent continuations.

B.4 Creative and Exploratory Decisions

Creative acts often occur at structural boundaries:

- Abandoning a nearly completed work
- Pursuing an unconventional idea
- Breaking from established paradigms

Here, the system departs from stable attractors and enters a region where:

- Prior structure weakens
 - Multiple trajectories become viable
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B.5 Cognitive Instability and Crisis States

Extreme psychological states may also approach non-uniqueness:

- Existential crises
- Decisions involving self-harm vs continuation
- Radical identity redefinition

These states are characterized by:

- Breakdown of stable internal models
 - Increased sensitivity to minimal perturbations
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B.6 Microphysical Hypothesis (Speculative)

Under hypotheses such as Diósi–Penrose or Orch-OR:

- Neural processes may involve non-computable dynamics
- State reduction may act as a selection mechanism

This suggests a possible physical substrate for selection over Δ , though no specific mechanism is assumed in this work.

B.7 Structural Interpretation

Across all examples, a common pattern emerges:

- The system approaches a boundary of structural stability
- Multiple trajectories become admissible
- A selection event resolves the multiplicity

Formally:

$$\text{Non-Lipschitz regime} \Rightarrow |\Delta| > 1 \Rightarrow \text{Selection}$$

B.8 Conceptual Conclusion

Human free will is not uniformly distributed across all decisions.

It is concentrated in:

critical boundary conditions where structural constraints fail to enforce a unique outcome.

These correspond, phenomenologically, to the most consequential decisions in a human life.

B.9 Summary Statement

Free will is exercised most strongly at structural boundaries, where the system admits multiple possible futures and must resolve them through selection.